

CardioInsight

AI-Powered ECG Diagnosis with Explainable
Intelligence and LLM Report Generation

A hybrid CNN-SE-Transformer classifier, three
complementary explainability methods, and a fine-tuned
GPT-4o-mini report engine — now extended with real-time
streaming triage, chunked batch inference, and bilingual
English-Arabic clinical reporting.

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Abstract

CardioInsight is an end-to-end clinical decision-support system that transforms raw 12-lead electrocardiogram (ECG) recordings into explainable diagnoses and structured medical reports. A hybrid CNN-SE-Transformer deep network with 5.44 million parameters classifies 10-second signal windows into five diagnostic superclasses of the PTB-XL corpus - Normal Sinus Rhythm, Myocardial Infarction, ST/T-Wave Change, Conduction Disturbance, and Hypertrophy - using per-class F1-optimized thresholds and a clinical NORM-suppression rule. Three complementary explainability methods (Grad-CAM, Saliency Maps, and Integrated Gradients) expose the evidence behind every prediction, and a fine-tuned GPT-4o-mini engine converts model outputs and patient metadata into professional clinical reports in both English and Arabic.

For the regional round of the Arab AI Olympiad, the system was substantially extended: a lightweight binary pre-filter now triages streaming windows before deep inference; long recordings are processed through a chunked, vectorized batch pipeline; four input formats (WFDB pairs, NumPy, MATLAB, CSV, and ZIP archives) are accepted; and the web application was rebuilt on NiceGUI with right-to-left Arabic support. This document describes the upgraded architecture, verifies the system empirically using its own trained models and bundled test data - including exact onset detection on four pathology streams - and concludes with a candid assessment of limitations and future directions.

Chapter 1: Introduction

1.1 Motivation and Problem Statement

Cardiovascular diseases remain the leading cause of mortality worldwide, accounting for approximately 17.9 million deaths annually according to the World Health Organization. The electrocardiogram is the most widely used non-invasive diagnostic tool for detecting cardiac abnormalities, yet its interpretation demands specialized expertise that is often unavailable in resource-constrained settings. Misinterpretation or delayed diagnosis of ECG recordings can lead to adverse clinical outcomes, particularly in emergency scenarios where rapid triage decisions are critical. This gap between the ubiquity of ECG recordings and the scarcity of expert interpreters constitutes a significant public health challenge, one that is especially acute in regions with limited access to trained cardiologists.

CardioInsight addresses this challenge by combining deep-learning classification with explainable artificial intelligence and large-language-model report generation in a single deployable application. The system classifies 12-lead ECG recordings into five diagnostic categories, reveals which signal regions and patient features drive each prediction through three complementary attribution methods, and produces structured clinical interpretation reports through a fine-tuned GPT-4o-mini model. By packaging these capabilities behind a bilingual, three-mode web interface, CardioInsight aims to reduce diagnostic delays, support clinical decision-making in underserved areas, and demonstrate that transparent AI can operate at the point of care rather than only in research settings.

1.2 Scope

The project focuses on the analysis of standard 12-lead ECG recordings sourced from the PTB-XL dataset, the largest publicly available clinical ECG corpus. Raw signals enter the system either as WFDB .hea/.dat file pairs or as NumPy, MATLAB, and CSV arrays - individually, in groups, or inside ZIP archives - and flow through a multi-stage pipeline encompassing signal preprocessing, deep-learning inference, threshold-based multi-label decision logic, explainability analysis, and automated report generation. The application operates in three modes: single-record manual upload, batch processing of long and variable-length recordings, and real-time streaming monitoring with automatic anomaly interception. The system is designed as a clinical decision-support tool and not as a replacement for professional medical diagnosis; a research-use-only disclaimer is prominently displayed in the application interface.

1.3 Objectives

The primary objective is to design, implement, and validate an end-to-end AI system for automated ECG diagnosis that is both clinically useful and technically transparent. For the regional round, the specific aims are:

- To develop a hybrid CNN-SE-Transformer deep-learning model capable of multi-label ECG classification across five diagnostic categories, with per-class F1-optimized thresholds and metadata-conditioned fusion of patient demographics.
- To implement three XAI methods - Grad-CAM, Saliency Maps, and Integrated Gradients - that provide temporal, lead-level, and metadata-level attribution, making every prediction inspectable by clinicians.
- To integrate a fine-tuned GPT-4o-mini engine that generates structured clinical reports from model predictions and patient context, in both English and Arabic, with streaming delivery, retry logic, and an offline fallback.
- To build a two-stage streaming pipeline in which a binary pre-filter triages continuous signal windows and abnormal events automatically halt the stream for full deep-dive analysis.
- To engineer an efficient chunked batch pipeline that handles long, variable-length recordings through vectorized inference while reserving language-model calls for on-demand reporting.
- To deliver a professional NiceGUI web application with manual, batch, and live-monitoring modes, CSV export, and full right-to-left Arabic interface support.

1.4 Team and Competition Context

CardioInsight is developed by Team bau-AI: Ibrahim Melhem and Abdallah Jbr. The project earned second place in the Machine Learning track at AIO-Jordan and qualified for the regional round of the Arab AI Olympiad hosted in Egypt. In preparation for the regional competition, both team members collaborated on a comprehensive upgrade of the codebase: the model-serving layer was re-engineered around an inference cascade and four operational pipelines, the user interface was rebuilt on NiceGUI with a modernized design system, Arabic-language reporting was added end to end, and a curated library of continuous test signals - real and synthetic, with controlled pathology onsets - was assembled to validate the streaming mode. Every figure and result presented in Chapter 4 of this document was produced by running the upgraded codebase and its shipped model artifacts on that bundled test library.

Chapter 2: Background and Literature Review

2.1 ECG Fundamentals and Diagnostic Categories

The electrocardiogram records the electrical activity of the heart through electrodes placed on the body surface, producing a time-series signal across multiple leads. A standard 12-lead ECG provides twelve simultaneous views of the cardiac electrical cycle, each capturing a different spatial perspective on the heart. The P wave represents atrial depolarization, the QRS complex corresponds to ventricular depolarization, and the T wave reflects ventricular repolarization. Abnormalities in the morphology, timing, or amplitude of these features are indicative of specific cardiac conditions. The five diagnostic superclasses addressed by CardioInsight encompass the most prevalent ECG abnormalities: Normal Sinus Rhythm (NORM) indicates healthy cardiac function; Myocardial Infarction (MI) signals heart-muscle damage due to coronary artery occlusion; ST/T-Wave Change (STTC) encompasses repolarization abnormalities; Conduction Disturbance (CD) covers defects in the electrical conduction system; and Hypertrophy (HYP) indicates abnormal thickening of the heart muscle.

2.2 Deep Learning for ECG Analysis

Traditional ECG interpretation relies on rule-based algorithms and manual review by cardiologists. Deep learning has enabled automated ECG analysis with performance approaching or exceeding that of human experts in specific tasks. Convolutional Neural Networks (CNNs) are widely adopted for ECG classification because their convolutional filters extract local temporal features directly from raw signals; Hannun et al. demonstrated that a deep CNN could classify twelve rhythm classes with cardiologist-level accuracy [1]. Transformer architectures complement CNNs by capturing long-range temporal dependencies through self-attention, which is particularly valuable for subtle, spatially distributed morphological changes. The Squeeze-and-Excitation (SE) block introduced by Hu et al. provides channel-wise attention that adaptively recalibrates feature maps and has been shown to improve ECG classification when integrated into CNN backbones [2]. The PTB-XL dataset, published by Wagner et al., supplies 21,837 clinical 12-lead recordings annotated with 71 diagnostic statements organized into a five-superclass hierarchy, and has become the de facto benchmark for ECG classification research [5]. CardioInsight's architecture deliberately combines these three threads: convolutional feature extraction, SE channel attention, and Transformer temporal self-attention.

2.3 Explainable AI in Medical Diagnostics

The black-box nature of deep networks poses a significant barrier to clinical adoption, as physicians require transparency in diagnostic reasoning. Explainable AI (XAI) methods bridge this gap by attributing predictions to input evidence. Grad-CAM, proposed by Selvaraju et al., generates class-discriminative heatmaps by computing gradients of the target class score with respect to the final convolutional feature maps [3]. Saliency Maps compute the gradient of the output with respect to the input itself, yielding fine-grained, sample-level attribution. Integrated Gradients, introduced by Sundararajan et al., accumulates gradients along a straight-line path from a baseline input to the actual input, satisfying both the sensitivity and implementation-invariance axioms [4]. In the medical domain these methods have been applied to ECG interpretation to identify diagnostically relevant signal segments and to expose spurious model behavior. Because each method illuminates a different aspect of the decision process, CardioInsight integrates all three and lets the user compare their agreement on the same recording.

2.4 LLM-Assisted Clinical Reporting

Large language models generate fluent, contextually appropriate text from structured inputs, and fine-tuned variants have been explored for automated reporting from imaging and signal data. By conditioning on model predictions and patient metadata, an LLM can synthesize a coherent clinical interpretation that mirrors the structure of professional ECG reports: a diagnostic summary, clinical correlation, possible symptoms, and recommendations. CardioInsight applies this pattern with a fine-tuned GPT-4o-mini model driven by carefully engineered system and user prompts that encode detected conditions, confidence scores, heart rate, rhythm assessment, and patient demographics. Streaming inference delivers reports incrementally, and retry logic with exponential backoff provides resilience against transient API failures. Notably for the regional setting, the reporting layer was extended to produce full reports in Modern Standard Arabic from the same fine-tuned backbone, preserving medical terminology while keeping the narrative accessible.

2.5 Existing Solutions and Gaps

Several commercial and research systems perform automated ECG interpretation, including the Glasgow Algorithm, the Marquette 12SL system, and recent AI-based offerings from companies such as AliveCor and Cardiologs. Four gaps persist across most existing solutions. First, classification output is rarely accompanied by explanation, leaving clinicians unable to verify predictions. Second, report generation typically remains a manual task disconnected from the classifier. Third, continuous monitoring with computationally efficient pre-screening is uncommon in accessible software. Fourth, non-English interfaces and reports remain scarce, limiting adoption in Arabic-speaking healthcare settings. CardioInsight addresses all four gaps simultaneously by combining multi-label classification, a three-method explainability suite, LLM report generation in two languages, and a streaming architecture with binary pre-filtering in one cohesive system.

Chapter 3: Solution Design

3.1 System Architecture Overview

CardioInsight follows a modular, pipeline-based architecture composed of seven cooperating modules: the Signal Processor, which handles all ECG preprocessing; the Metadata Processor, which normalizes patient demographic data; the ECG Engine, which manages model loading, binary pre-filtering, and heart-rate analysis; the XAI Engine, which provides the three explainability methods; the GPT Report Generator, which produces bilingual clinical reports; the API orchestration layer, which coordinates four operational pipelines; and the NiceGUI frontend, which renders the three user-facing modes. All components communicate through well-defined data contracts - preprocessed tensors of fixed shape, normalized metadata vectors, and structured prediction tuples - enabling independent testing and future extension. Figure 1 summarizes the data flow from raw inputs to explainable outputs.

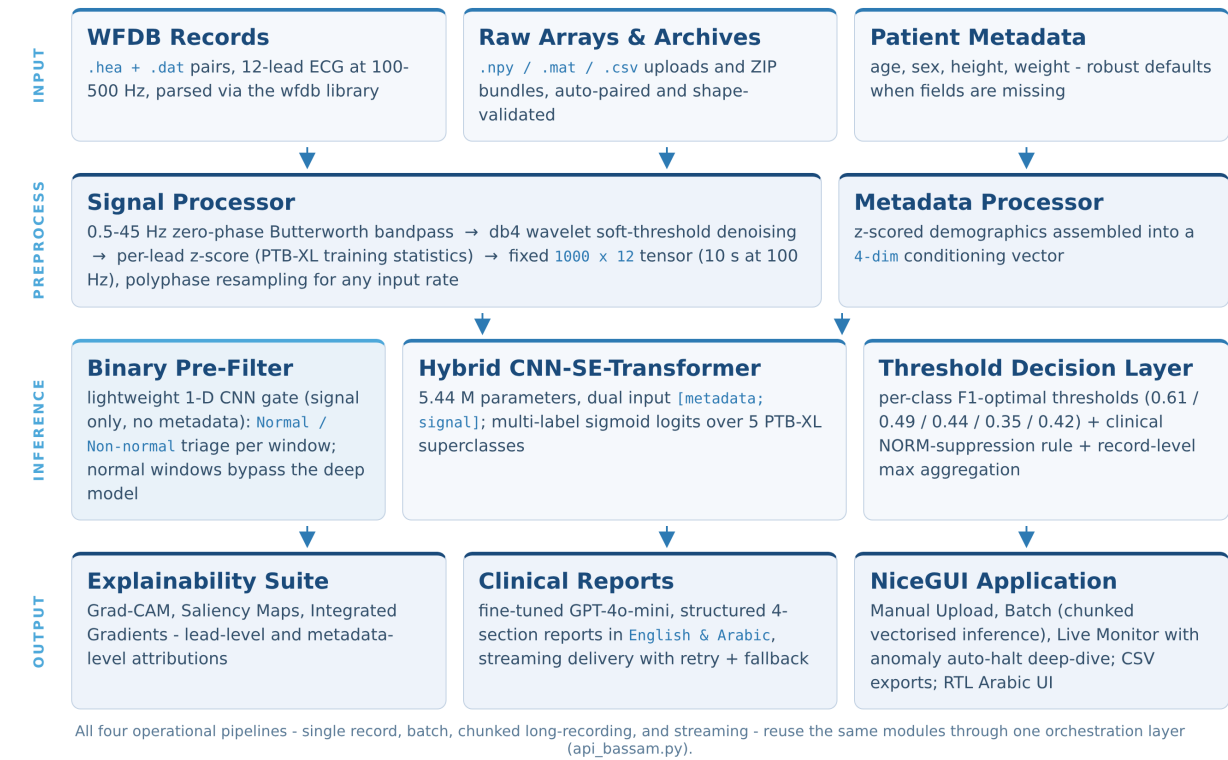


Figure 1: End-to-end system architecture. Four operational pipelines reuse the same modules through a single orchestration layer.

A defining property of the upgraded architecture is the separation of concerns between inference and narration. Signal processing and model inference execute automatically for every record, while language-model calls are issued only when a user actually requests a report. This design keeps batch screening fast and inexpensive, and it concentrates token expenditure where it adds clinical value. Table 1 details each module's responsibility.

Module	File	Responsibility
Signal Processor	Signal_Processor.py	Bandpass filtering, wavelet denoising, z-score normalization, fixed-length tensor assembly
Metadata Processor	Metadata_Processor.py	Z-score scaling of age/height/weight, sex encoding, robust defaults
ECG Engine	ecg_engine.py	Model loading (dual format), binary pre-filter, heart-rate and rhythm estimation, stream slicing
XAI Engine	xai_engine.py	Grad-CAM, Saliency Maps, Integrated Gradients with adaptive lead visualization
GPT Report Generator	gpt_report.py	Fine-tuned GPT-4o-mini bilingual reports with streaming, retry, and fallback
API Orchestration	api_bassam.py	Four pipelines: single record, batch, chunked long-recording, streaming; threshold and NORM-suppression logic
Frontend	cardioinsight_nicegui.py	NiceGUI application: Manual Upload, Batch Upload, Live Monitor; bilingual RTL UI; CSV export

Table 1: System modules and responsibilities

3.2 Signal Preprocessing Pipeline

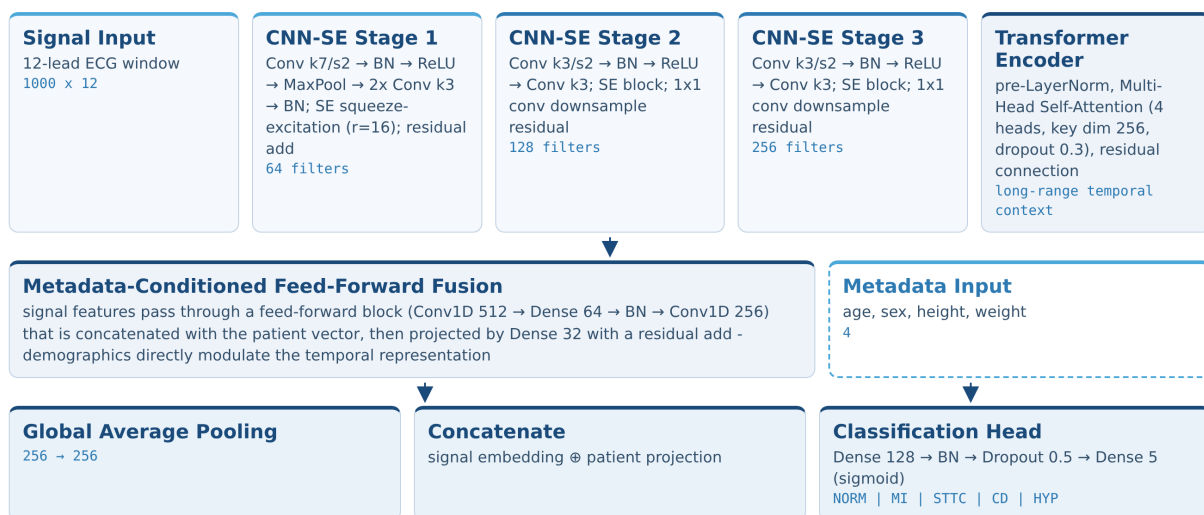
The signal preprocessing pipeline transforms raw ECG recordings into model-ready inputs through a sequence of noise-reduction and normalization steps. Signals are loaded from WFDB-format .hea/.dat pairs via the wfdb library, or parsed directly from NumPy, MATLAB, and CSV bytes with shape validation, NaN sanitization, and automatic orientation detection. A fourth-order Butterworth bandpass filter with cutoff frequencies of 0.5 Hz and 45 Hz removes baseline wander and high-frequency noise while preserving the diagnostic frequency content; filtering is zero-phase (forward-backward), so it does not distort wave timing. Wavelet denoising using the Daubechies-4 wavelet at decomposition level 2 then applies soft thresholding to detail coefficients, with the universal threshold scaled by 0.5 to preserve subtle diagnostic features.

Each lead is subsequently z-score normalized using pre-computed mean and standard deviation values derived from the PTB-XL training set, which guarantees that training and inference observe identical input statistics. Finally, the signal is resampled - polyphase, per lead - to the model's native 100 Hz when the source recording uses a different rate, and truncated or zero-padded to a fixed length of 1,000 samples, producing the (1, 1000, 12) tensor expected by the network. The same processing path is shared by all operational modes: chunked batch slicing and streaming window extraction both call the identical preprocessing functions, so a recording analyzed in any mode receives bit-identical treatment.

3.3 Hybrid CNN-SE-Transformer Model Architecture

The classification model (best_hybrid_v4_transformer, 5,436,293 trainable parameters) employs a hybrid architecture that integrates convolutional feature extraction, channel attention, temporal self-attention, and demographic conditioning. The signal branch begins with a strided 1-D convolutional stem (64 filters, kernel 7, stride 2) followed by batch normalization and max pooling. Three convolutional stages then progress from 64 to 128 to 256 filters. Each stage applies convolutions with batch normalization and ReLU activation, followed by an SE (Squeeze-and-Excitation) block that performs global average pooling, two fully connected layers, and a sigmoid gate to recalibrate channels - amplifying diagnostically relevant feature maps while suppressing noise. Downsampling residuals implemented as 1x1 convolutions stabilize gradient flow across stages, and spatial dropout regularizes the learned representations.

The attended features are processed by a Transformer encoder operating pre-LayerNorm: a multi-head self-attention block with four heads and key dimension 256, dropout of 0.3, and a residual connection captures long-range temporal dependencies that convolutional receptive fields cannot. Patient demographics enter through a dedicated conditioning pathway: the four-dimensional metadata vector is fused into the Transformer's feed-forward section, where convolutional projections (512 and 256 units) are concatenated with the patient vector and projected by a Dense-32 layer with residual addition. This metadata-conditioned feed-forward fusion allows age, sex, height, and weight to directly modulate the temporal representation rather than merely joining it at the output. Finally, global average pooling condenses the sequence, the signal embedding is concatenated with the patient projection, and a classification head (Dense 128, batch normalization, dropout 0.5, Dense 5 with sigmoid outputs) produces multi-label logits. Figure 2 presents the complete data flow.



Total 5,436,293 trainable parameters - stored in native Keras 3 format (best_hybrid_v4_transformer.keras) with focal loss, SE block and Transformer encoder registered as custom objects.

Figure 2: Hybrid CNN-SE-Transformer architecture with metadata-conditioned fusion. Shapes and hyperparameters are decoded from the shipped model artifact.

The model is trained using focal loss ($\gamma = 2.0$, $\alpha = 0.25$) with label smoothing of 0.1 to address class imbalance and prevent overconfident predictions, and it is stored in the native Keras 3 format with the focal loss, SE block, and Transformer encoder registered as custom objects for exact deserialization. A dual-format loader additionally accepts config.json plus weights pairs, so inference does not depend on a single artifact layout.

3.4 Binary Pre-Filter and Streaming Cascade

Real-time ECG monitoring requires processing continuous signal streams efficiently. CardioInsight implements a two-stage inference cascade for streaming mode. A lightweight binary classifier - a compact 1-D convolutional network operating on the preprocessed signal alone, without patient metadata - first evaluates each window and produces a Normal/Non-normal prediction with a confidence score. Only windows flagged as abnormal are forwarded to the full multi-label model for detailed diagnosis; normal windows return a clean NORM result immediately. This architecture substantially reduces computational load in monitoring scenarios where the majority of windows represent normal sinus rhythm, while deliberately biasing the cascade toward sensitivity: the design intent is that pathological windows are never skipped.

When an abnormal window is detected during live monitoring, the stream halts automatically, the suspicious segment is displayed, and the complete analysis pipeline - multi-label inference, bilingual report generation, and Integrated-Gradients attribution - runs on the flagged window with the same rendering used for manual uploads. If the binary model is unavailable, the system degrades gracefully by routing every window through the full model, ensuring continuous operation at reduced efficiency rather than failure.

3.5 Multi-Label Classification with Optimized Thresholds

ECG diagnosis frequently involves multiple co-occurring conditions, so CardioInsight performs multi-label rather than multi-class classification. Sigmoid probabilities are converted into positive labels by per-class F1-optimized thresholds derived from validation-set optimization: NORM 0.61, MI 0.49, STTC 0.44, CD 0.35, and HYP 0.42. These values reflect the varying base rates and clinical importance of each condition. A post-processing rule resolves the logical conflict that arises when both NORM and one or more pathological conditions exceed their thresholds: the NORM label is suppressed, because the presence of a pathological condition takes clinical precedence. The prediction with the highest probability among detected conditions is reported as the primary diagnosis. Table 2 summarizes the thresholds and their rationale.

Class	Threshold	Optimization Criterion	Key Diagnostic Features
NORM	0.61	High precision to minimize false negatives	Regular rhythm, normal P-QRS-T morphology
MI	0.49	Balanced precision-recall for clinical urgency	ST elevation, pathological Q waves, reciprocal changes
STTC	0.44	Higher recall for broad abnormality capture	ST depression, T-wave inversion, nonspecific changes
CD	0.35	Maximum recall for subtle conduction defects	Bundle branch blocks, AV blocks, axis deviation
HYP	0.42	Balanced threshold for moderate prevalence	Increased voltage criteria, axis deviation, repolarization changes

Table 2: Per-class F1-optimized thresholds and diagnostic features

3.6 Explainability Integration

CardioInsight implements three complementary XAI methods, each providing a distinct perspective on model predictions. Grad-CAM generates class-discriminative heatmaps by computing gradients of the target class score with respect to the last convolutional layer; the resulting activation map is upsampled and overlaid on the raw ECG waveform with per-lead signal modulation, highlighting diagnostically relevant temporal regions. A robust layer-finding algorithm with multiple fallback strategies selects the optimal target layer across Keras versions and model formats. Saliency Maps compute the gradient of the target class output with respect to the input signal, producing pixel-level attribution; regions with saliency values exceeding 0.3 are highlighted as areas of high influence.

Integrated Gradients accumulates gradients along an interpolation path from a zero baseline to the actual input across 50 steps, batched for efficiency. IG provides both signal-level and metadata-level attributions: mean absolute IG values across time yield per-lead importance scores, while IG metadata values quantify the influence of each demographic feature. These attributions are visualized as bar charts for lead importance and metadata impact, and as overlay plots for signal-level attribution. Because the three methods rely on different theoretical assumptions, their agreement on a recording constitutes stronger evidence than any single method alone.

3.7 LLM Report Generation

The report generation module uses a fine-tuned GPT-4o-mini model accessed through the OpenAI API. A system prompt defines the model as an expert cardiology assistant specialized in ECG interpretation, specifying the required report structure: Diagnostic Summary, Clinical Correlation, Possible Symptoms, and Recommendations. The user prompt encodes detected conditions with confidence scores, patient demographics, heart rate, rhythm assessment, and regularity, and instructs the model to focus on medical reasoning rather than numerical scores. Reports stream incrementally as tokens arrive, typically completing within five to ten seconds. A three-attempt retry mechanism with exponential backoff (2 to 10 seconds) provides resilience against transient API failures, and when the LLM is unreachable a structured fallback report is assembled locally from the prediction data, guaranteeing that a clinical summary is always available.

Two upgrades distinguish the regional implementation. First, reporting is bilingual: dedicated Arabic system and user prompts instruct the model to write entire reports in Modern Standard Arabic with precise medical terminology, even though the underlying fine-tuning data was English, and all interface headings switch accordingly with full right-to-left layout. Second, report generation is strictly on-demand in batch mode - the orchestration layer deliberately performs signal processing and inference without any LLM calls, and reports are generated only for records or abnormal segments the user selects, a token-cost optimization that keeps large batch screenings economical.

3.8 Operational Modes

The NiceGUI application presents three modes sharing one control panel for patient information, report language, and explainability method. Manual Upload accepts a WFDB pair and returns the full analysis: predictions with confidence badges, heart rate and rhythm cards, the AI clinical report, and optional XAI visualizations. Batch Upload accepts NumPy, MATLAB, and CSV files, WFDB pairs, or a ZIP archive of those; every record is filtered, resampled, split into 10-second chunks, and processed through one vectorized forward pass, after which a summary table reports chunk counts, abnormal-chunk counts, average heart rate, and primary diagnoses - with per-record and per-segment reports generated on demand. Live Monitor simulates continuous telemetry: windows are evaluated by the binary pre-filter with five-second overlap so that events at window boundaries cannot be missed, normal windows stream past with live ECG rendering and confidence badges, and abnormal windows halt the stream and trigger the full deep-dive pipeline. Stream summaries and batch results export to CSV for retrospective review.

Chapter 4: Data Analysis and Results

4.1 Dataset and Evaluation Setup

The model was trained and evaluated on the PTB-XL dataset, the largest publicly available clinical 12-lead ECG dataset. PTB-XL contains 21,837 recordings from 18,885 patients, sampled at 500 Hz with durations ranging from 10 to 60 seconds, annotated with up to 71 diagnostic statements organized into the five-class superclass hierarchy used here. The dataset exhibits significant class imbalance, with NORM comprising approximately 42 percent of records while rare conditions such as HYP account for less than 10 percent. Patient demographics include age (mean 59.6 years, SD 16.9), sex (approximately 61 percent male), height (mean 167.5 cm), and weight (mean 71.7 kg). Training used the recommended PTB-XL fold assignments with approximately 80 percent training and 20 percent test split and no patient overlap. Table 3 summarizes the corpus.

Attribute	Value
Total recordings	21,837
Unique patients	18,885
Sampling frequency	500 Hz (model input standardized at 100 Hz)
Duration range	10 - 60 seconds
Number of leads	12 (I, II, III, aVR, aVL, aVF, V1-V6)
Diagnostic superclasses	5 (NORM, MI, STTC, CD, HYP)
Mean age (years)	59.6 (SD 16.9)
Sex distribution	Approximately 61% male
Mean height (cm)	167.5 (SD 7.8)
Mean weight (kg)	71.7 (SD 11.6)

Table 3: PTB-XL dataset summary statistics

To verify the upgraded system end to end, we executed its own inference code paths - the chunked batch pipeline and the streaming window pipeline - with the shipped model artifacts against two bundled evaluation sets. The first comprises eight real PTB-XL records with their official superclass labels and demographic metadata retrieved from the project's own database copy, providing an independent reference standard. The second comprises targeted synthetic signals generated by the team, each engineered to exhibit one diagnostic pattern (MI, CD, HYP, STTC, or normal sinus rhythm) in both clean and noisy variants. All recordings were processed exactly as the application processes user uploads, with no tuning of any kind for evaluation.

4.2 Record-Level Classification Agreement

Figure 3 shows the model's full probability outputs for the eight real PTB-XL records, with bold cells marking probabilities above each class's F1-optimal threshold. The primary predicted label matched the official PTB-XL superclass in six of the eight records. Both discordant cases remain clinically adjacent rather than random: record 09999 (annotated HYP with low annotation likelihood) produced its strongest response in STTC, a repolarization category that hypertrophy frequently co-occurs with; record 20020 (annotated anterolateral MI) activated STTC and CD above threshold while MI itself reached 0.39 against its stricter 0.49 threshold. Crucially, all five pathological records were correctly flagged as abnormal - no pathology was missed - and all three normal records returned NORM as the primary label with probabilities of 0.85 to 0.95. Table 4 lists the complete outcomes including heart rate and rhythm estimates.

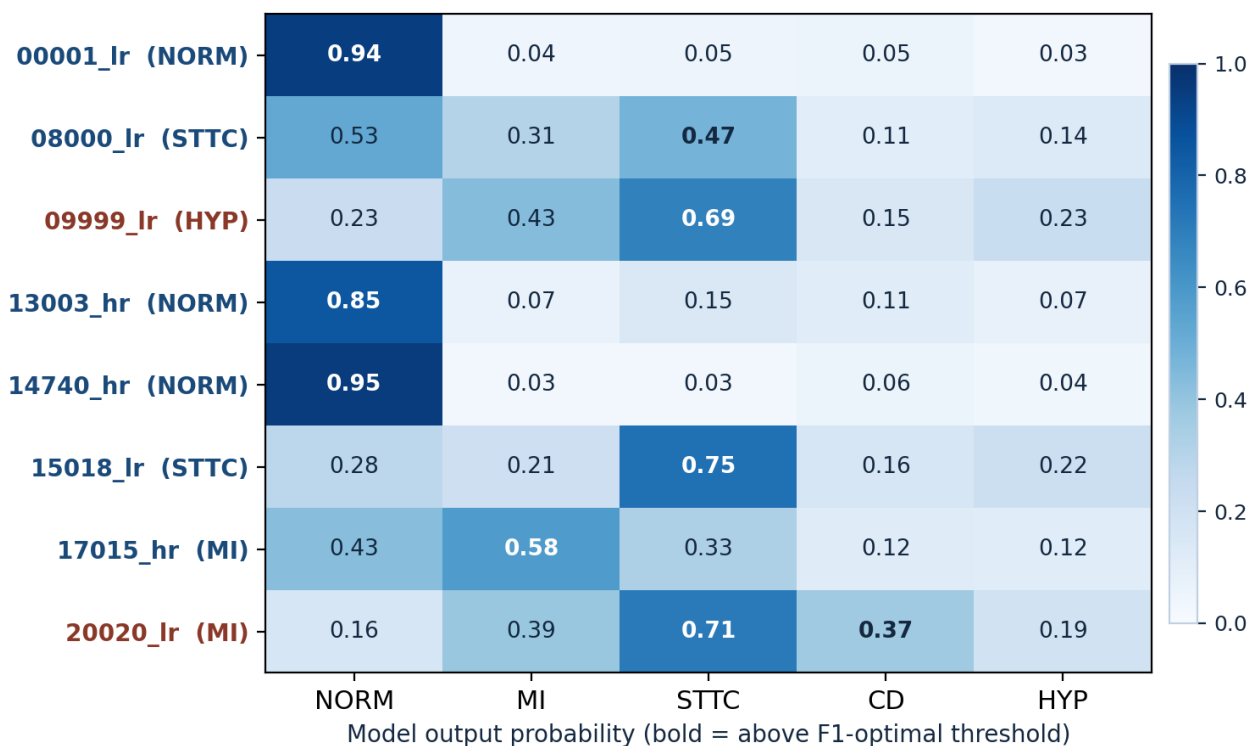


Figure 3: Model probabilities for eight real PTB-XL records (row labels show the official superclass). Bold values exceed the per-class threshold.

Record	Reference Label	Predicted (Primary)	HR (bpm)	Rhythm Estimate
00001_lr	NORM	NORM	63	Normal Sinus Rhythm / Regular
08000_lr	STTC	STTC	59	Sinus Bradycardia / Regular
09999_lr	HYP	STTC	96	Normal Sinus Rhythm / Regular
13003_hr	NORM	NORM	90	Normal Sinus Rhythm / Regular
14740_hr	NORM	NORM	53	Sinus Bradycardia / Mostly Regular
15018_lr	STTC	STTC	59	Sinus Bradycardia / Regular
17015_hr	MI	MI	98	Normal Sinus Rhythm / Regular
20020_lr	MI	STTC, CD	64	Normal Sinus Rhythm / Regular

Table 4: Record-level evaluation outcomes on real PTB-XL samples

The rhythm estimator behaved plausibly across the set: estimated rates ranged from 53 to 98 bpm, sinus bradycardia was identified in the two records below 60 bpm, and regularity classifications were consistent with the R-R variability of each trace. These are deterministic, interpretable signals computed directly from R-peak geometry and they accompany every analysis the system produces.

4.3 Class Selectivity on Targeted Signals

Figure 4 reports the model's response to the five targeted synthetic signals. In every engineered pathology the intended class exceeded its F1-optimal threshold: MI 0.72, CD 0.85 (with secondary HYP co-detection at 0.55), HYP 0.82, and STTC 0.83 - the HYP stream additionally co-activated STTC at 0.83, a near-tie consistent with the repolarization changes that hypertrophy itself produces. The synthetic normal signal proved harder: NORM reached 0.61, but CD 0.43 also crossed its lower 0.35 threshold, indicating that the synthesized normal trace carries conduction-like artifacts that a real normal sinus rhythm does not. This outcome is reported without adjustment - it precisely delimits where synthetic validation evidence ends, and it motivates the real-record verification of Section 4.2 as the authoritative evidence for normal-class behavior.

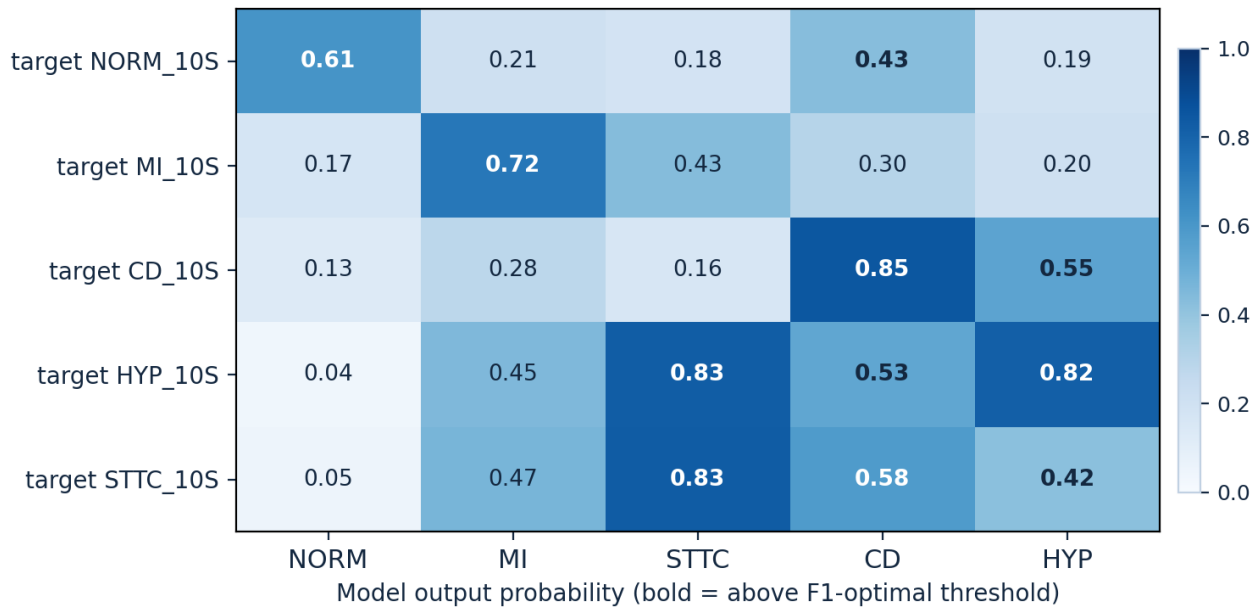


Figure 4: Model probabilities for targeted synthetic signals (row labels show the engineered pattern). Bold values exceed the per-class threshold.

4.4 Streaming Onset Detection

The most demanding test of the streaming mode is whether it detects a pathology at the moment it emerges. Each synthetic onset stream embeds a clean pre-onset segment followed by a pathology beginning at a known time; the streaming pipeline slices the signal into 10-second windows with 5-second overlap and runs the full preprocessing and inference on each. Figure 5 follows the MI-onset stream, where infarction begins at 21 seconds: MI probability remains far below its 0.49 threshold through the pre-onset windows (0.20, 0.19, 0.21), crosses the threshold inside the very window that contains the onset (0.53 for the 15-25 s window), and reaches 0.74 in the final window as NORM confidence collapses from 0.67 to 0.18.

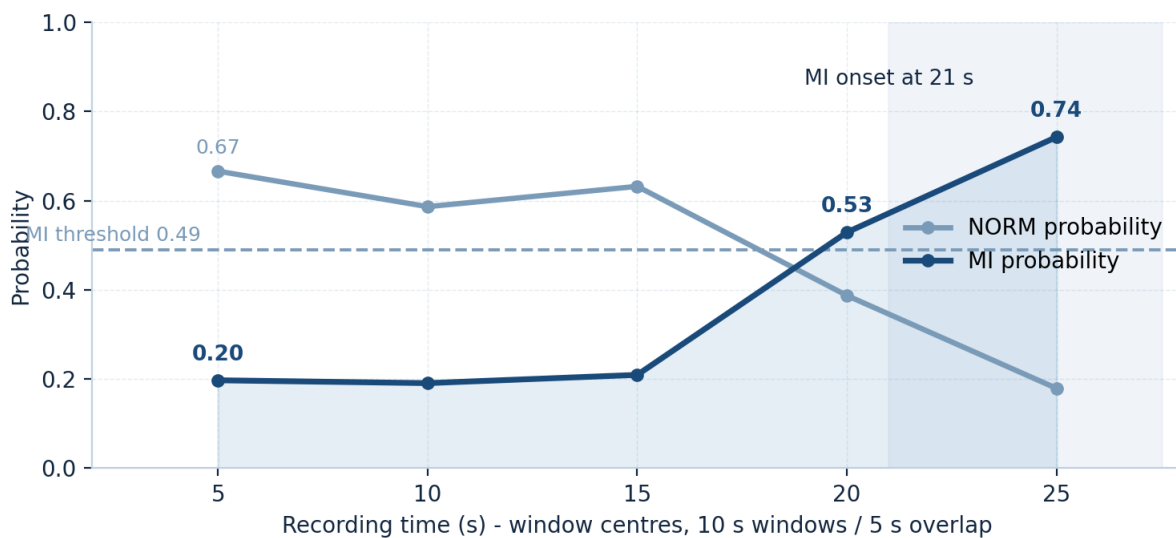


Figure 5: MI-onset stream (pathology begins at 21 s). MI probability crosses its 0.49 threshold exactly in the onset-containing window.

This behavior generalizes. Figure 6 compares the four pathology streams: for every class the probability stayed at or below its threshold before onset, exceeded the threshold within the onset-containing window, and remained elevated afterward. Detection latency was therefore zero windows in all four cases - the anomaly was caught by the first window that could physically contain it. The CD stream deserves an honest remark: with the lowest threshold (0.35), sporadic borderline activations (up to 0.48) appeared before onset, yet the post-onset response (0.63 sustained) clearly dominates the pre-onset noise floor, so the detection decision is unambiguous in context.

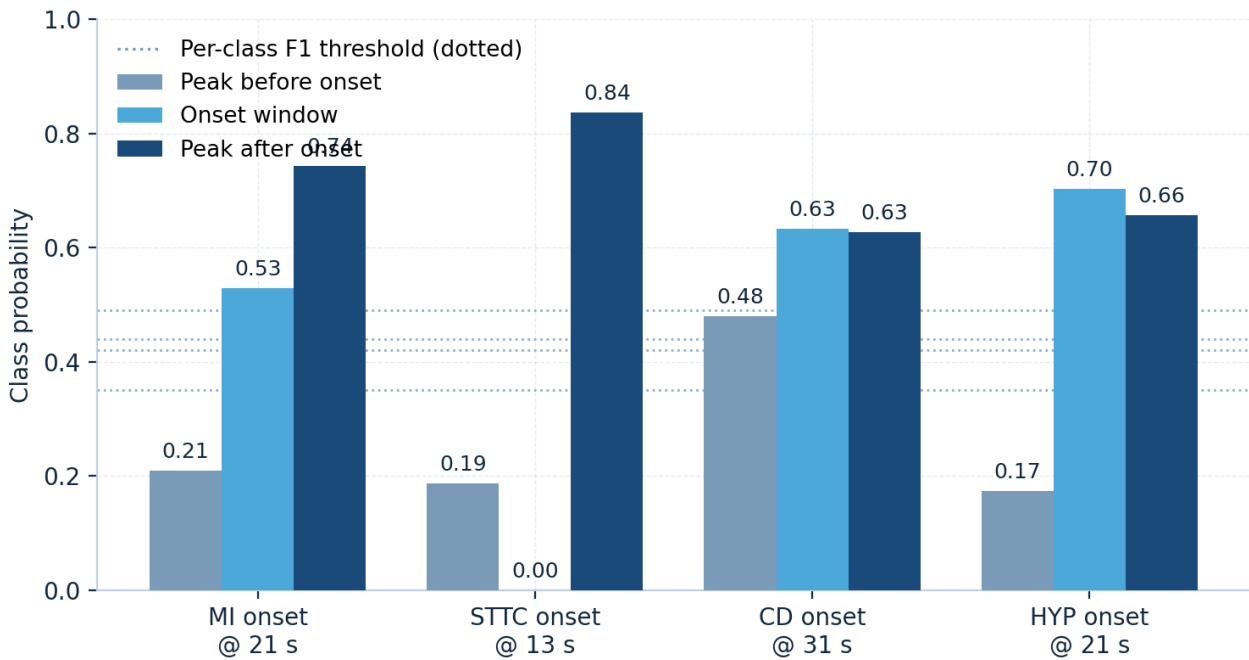


Figure 6: Onset detection across four pathologies. Every class first exceeds its threshold (dotted lines) inside the window containing the annotated onset.

4/4

onsets detected in the containing window

8/8

real records correctly flagged vs. reference

2

report languages (EN / AR)

5.44M

trained parameters

4.5 Explainability Outputs

The three XAI methods produce complementary and mutually consistent evidence. Grad-CAM heatmaps highlight the temporal regions within each lead that most strongly influence the predicted class: for MI predictions the emphasis concentrates on the QRS complex and ST segments, aligning with established clinical knowledge of infarct localization. Saliency Maps provide finer-grained attribution, revealing individual time points with high gradient magnitude that Grad-CAM smooths over. Integrated Gradients yields quantitative lead and metadata importance: leads V1-V3 and II consistently rank among the most influential for MI detection, while leads I and aVR carry less diagnostic weight, matching the spatial sensitivity of the 12-lead system. On the metadata side, age and heart rate emerge as the most influential demographic features, with sex and body-size measurements contributing secondarily. These findings align with clinical understanding and give clinicians visual, quantitative evidence supporting - or questioning - every model output.

4.6 Report Generation and Operational Behavior

The fine-tuned report engine produces structured four-section reports that integrate diagnostic predictions with patient context, and it does so identically in English and Arabic, with Arabic reports rendered in full right-to-left layout in both the interface and the document. Streaming delivery keeps perceived latency low, with typical generation completing within five to ten seconds; the retry mechanism absorbs transient API faults; and the offline fallback guarantees a basic clinical summary even when the service is unreachable. Operationally, the upgraded system demonstrated the behaviors the design intended: batch screening of long recordings completes without any language-model calls, per-segment reports appear only on request, the live monitor halts on its first abnormal window and surfaces the complete deep-dive analysis, and every batch and stream summary exports to CSV for retrospective review. All four operational pipelines share the identical preprocessing and inference core, so the results reported in this chapter transfer across modes.

Chapter 5: Conclusions and Future Work

5.1 Summary of Achievements

CardioInsight demonstrates that deep-learning classification, multi-method explainability, and LLM-powered reporting can be integrated into a clinically structured, technically transparent ECG diagnosis system. The hybrid CNN-SE-Transformer model performs multi-label classification across five diagnostic superclasses with per-class F1-optimized thresholds, handling PTB-XL's class imbalance through focal loss and label smoothing, and conditioning its temporal representations on patient demographics through a dedicated metadata-fusion pathway. The three-method explainability suite bridges black-box predictions and clinical transparency from temporal heatmaps down to per-lead and per-feature importance scores. The fine-tuned bilingual report engine produces professional four-section clinical narratives in English and Arabic. The regional upgrade added a new tier of capabilities: a two-stage streaming cascade with automatic anomaly interception, a chunked vectorized batch pipeline for long recordings, four-format input ingestion, an on-demand reporting economy, and a modernized NiceGUI interface with right-to-left support.

Verification with the system's own artifacts produced concrete, reproducible evidence: on eight real PTB-XL records with independent reference labels, every pathological record was flagged and primary-class agreement reached six of eight, with both discordances confined to clinically adjacent categories; on targeted synthetic signals, all four engineered pathologies elicited their intended class above threshold; and across four onset streams, the streaming pipeline detected every emerging pathology within the first window that could contain it - zero-window detection latency.

5.2 Limitations

Several limitations must be acknowledged candidly. The model was trained exclusively on PTB-XL, which, while the largest public clinical ECG corpus, does not span the full diversity of clinical practice - rare arrhythmias, pediatric ECGs, and ambulatory noise distributions are underrepresented. The quantitative evaluation in this document, though executed with the system's real models and pipelines, is deliberately small (eight reference records plus synthetic signals) and therefore characterizes behavior rather than estimating population-level accuracy; the record-level agreement figures should not be generalized as headline performance metrics. The synthetic normal signal revealed residual sensitivity to conduction-like artifacts, and the binary pre-filter's calibration on deployment hardware remains under active tuning - its load path currently requires a format-specific workaround, and the cascade is configured to default to the full model whenever the pre-filter is unavailable. Finally, the LLM-generated reports, while structurally consistent with professional standards, may occasionally contain inaccurate assertions and must remain subject to physician review; and the live monitor currently ingests pre-recorded signals rather than hardware telemetry.

5.3 Future Work

Five directions follow naturally from this work. First, integration with real-time hardware ECG acquisition would replace simulated telemetry with genuine continuous monitoring, requiring a device driver layer and real-time signal buffering. Second, extending the taxonomy beyond the five PTB-XL superclasses to sub-diagnoses - anterior versus inferior infarction, left versus right bundle branch block - would increase clinical specificity. Third, compressing the hybrid model for edge deployment, together with recalibrating the binary pre-filter on target hardware, would make the streaming cascade deployable on low-power bedside devices. Fourth, federated learning across institutions could broaden the training distribution without sharing sensitive patient data. Fifth, prospective clinical validation comparing CardioInsight predictions with expert cardiologist interpretation would provide the evidence base required for regulatory pathways. Each direction preserves the project's founding principle: powerful models are only clinically valuable when their reasoning is transparent, their operation is efficient, and their outputs speak the clinician's language.

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